Jonghyun Shin

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About

I am a second-year M.S.-Ph.D. integrated student in Artificial Intelligence at **Korea University**, advised by **Prof. Sejun Park**. My research studies how finite numerical precision and finite randomness affect the theoretical behavior of machine learning algorithms, with particular interests in **generalization**, **algorithmic stability**, and **differential privacy**.

Education

Korea University — *Seoul, South Korea*
M.S.-Ph.D. Integrated Program in Artificial Intelligence — *Sep. 2025 – Present*
- GPA: 4.5/4.5; advised by Prof. Sejun Park.

Korea University — *Seoul, South Korea*
B.S. in Mathematics Education / Statistics (Double Major) — *Mar. 2019 – Aug. 2025*
- GPA: 4.31/4.5.

Gyeongnam Science High School (GSHS) — *Gyeongsangnam-do, South Korea*
Mar. 2017 – Dec. 2019

Publications & Preprints

International Conference

[C1] Jonghyun Shin, Namjun Kim, Geonho Hwang, Sejun Park, Minimum Width for Universal Approximation using Squashable Activation Functions, Proceedings of the International Conference on Machine Learning (ICML), 2025 [arXiv]

Preprint

[P1] Jonghyun Shin, Sejun Park, Uniform Stability and Generalization Error of GD and SGD on Fixed-Point Parameters, [arXiv]

Research Experience

Efficient Inference and Learning Lab — *Korea University, Seoul*
Undergraduate Research Student — *Dec. 2023 – Aug. 2025*
- Advised by Prof. Sejun Park.
- Research topic: characterizing minimum width for universal approximation using general activation functions.

Awards & Honors

Master's Excellence Scholarship in Science and Engineering — *Sep. 2025 – Aug. 2027*
Korea Student Aid Foundation

Dean's List — *Mar. 2021, Sep. 2023*
Korea University

Teaching & Academic Service

Teaching Assistant, Korea University — *2025*
Discrete Mathematics (Spring 2025); Calculus (Fall 2025)

Teaching Practicum, Daeshin High School — *Apr. 2025*
Seoul, South Korea

Reviewer, NeurIPS — *2026*

Skills

Languages: Korean (native), English | **Programming:** Python, R | **Tools:** LaTeX